Applied Linear Algebra in Data Analysis

Case Study 01: Clustering, Classification, and Regression

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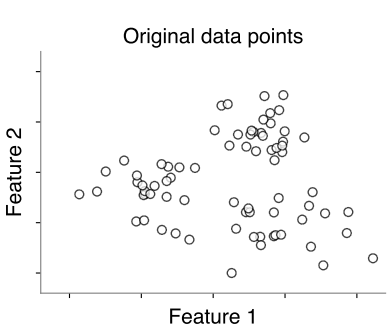
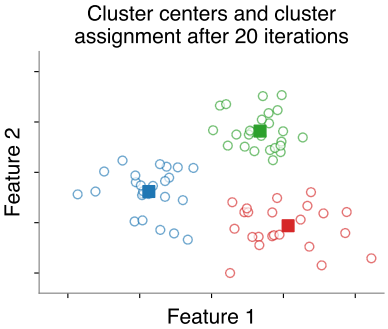
Clustering of data points

Clustering: Grouping similar items together.

- There are various clustering algorithms: k-means, hierarchical clustering, Gaussian mixture models, etc.
- **k-means** is the simplest and most popular clustering algorithm.





k-means clustering

- The k-means algorithm is an iterative algorithm that divides a group of N samples (n-vectors) into k clusters.
- Clustering is done by minimizing the following cost,

$$J_{clust} = \frac{1}{N} \sum_{j=1}^{k} \sum_{i \in C_j} \|\mathbf{x}_i - \mathbf{m}_j\|_2^2$$

- Minimizing this cost for a given dataset is computationally intensive, because the optimal choice for the means $\mathbf{m}_j$ and the cluster assignments C_j depend on each other.
- k-means takes a simpler approach: minimizing J_{clust} when either the means or the cluster assignments are fixed is easy.

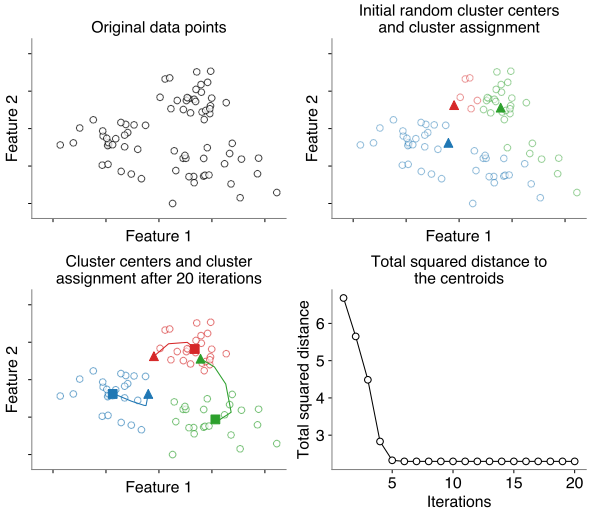
k-means clustering

- k-means solves the clustering problem by alternately fixing the means and the cluster assignments, while updating the other.

- k-means has two steps: we first randomly choose some cluster means.
  - **Cluster assignment update**: For a fixed set of cluster means, find cluster assignments that minimize J_{clust}.

  - **Cluster means update**: For a fixed cluster assignment, find the means $\mathbf{m}_i$ that minimize J_{clust}.
- Applying these two steps one after the other will lead to the algorithm converging towards a set of cluster means and assignments, because J_{clust} is guaranteed to reduce with each step.

k-means clustering



Classification and regression problems

- Both **classification** and **regression** involve finding a mapping from a feature vector $\mathbf{x} \in \mathbb{R}^n$ to a target value y.

- **Classification**: The target y takes values from a finite set of class labels, $y \in \{1, 2, \ldots, L\}$.

- **Regression**: The target y is a real-valued scalar, $y \in \mathbb{R}$.

- In both cases, we are given a *training dataset* $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ and use it to predict the target for new, unseen feature vectors $\mathbf{x}_{new}$.

k-nearest neighbors (k-NN) classifier

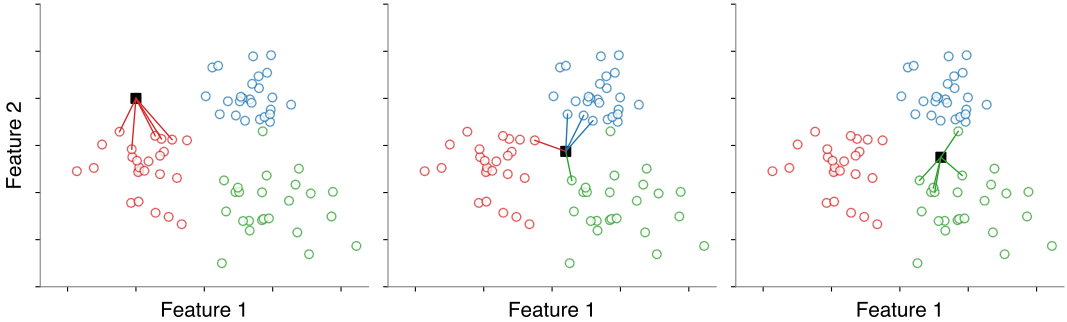
k-NN classifier: Classify a new point by majority vote of its k nearest neighbors in the training data.

- Given training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with $\mathbf{x}_i \in \mathbb{R}^n$ and labels $y_i \in \{1, \ldots, L\}$.
- For a new point $\mathbf{x}_{new}$, compute the Euclidean distance $\|\mathbf{x}_i - \mathbf{x}_{new}\|_2$ to every training sample.
- Select the k closest training samples; let $\mathcal{K}$ denote the index set of these k nearest neighbors.
- Assign the label with the highest frequency among $\{y_i\}_{i \in \mathcal{K}}$ as the predicted label $\hat{y}_{new}$.
- The only hyperparameter is k. There is no explicit training – the entire dataset is stored and used at prediction time.

k-NN classifier

k-NN Classifier Example

Consider an example where we have a collection of points in $\mathbb{R}^2$ belonging to three different classes. We wish to classify a new point from $\mathbb{R}^2$ as belonging to one of these three classes. Here we have chosen $k = 5$.



k-nearest neighbors (k-NN) regressor

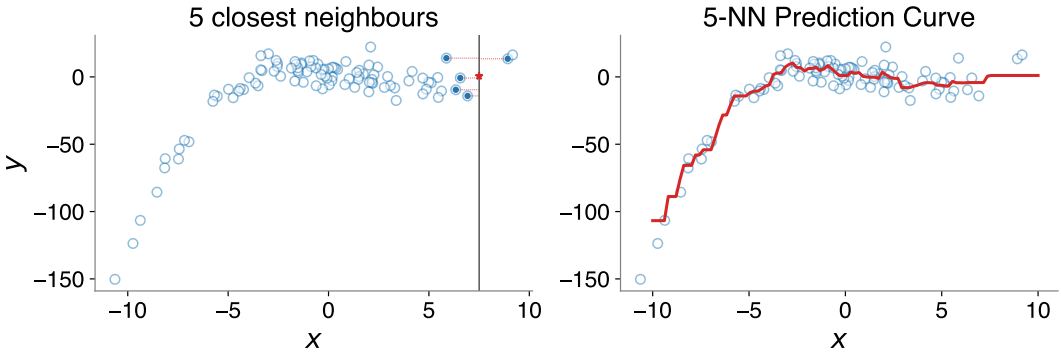
k-NN regressor: Same principle as the k-NN classifier, but the prediction is the *average* of the target values of the k nearest neighbors.

- Given training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$.
- For a new point $\mathbf{x}_{new}$, find the index set $\mathcal{K}$ of the k nearest neighbors using Euclidean distance.
- The predicted value is the mean of the neighbors' target values:

$$\hat{y}_{new} = \frac{1}{k} \sum_{i \in \mathcal{K}} y_i$$

- The key difference from the classifier: instead of a majority vote, we compute an average.

k-NN regressor

k-NN Regressor Example

Consider an example where we have a collection of scalar observations $\{(x_i, y_i)\}_{i=1}^{N}$. For a given new x_{new}, we wish to predict the corresponding y_{new}. Here we have chosen $k = 5$.